Table 6. Intended-use-specific quality and regulatory considerations for salivary gland organoids and derived products

Application	Quality and evidence priorities	Regulatory considerations
Research-grade organoids	Document tissue source, donor and species, culture conditions, passage history, cellular composition, contamination controls and assay reproducibility.	Standardization supports reproducible research but does not establish clinical-grade suitability.
PDO drug-sensitivity testing	Establish tumor authenticity, analytical performance, clinically relevant assay turnaround, clinical validity and clinical utility.	In the US, CLIA governs clinical laboratory quality; the FDA's 2024 LDT rule was vacated in 2025. In the EU, IVDR requirements and the conditional Article 5(5) exemption depend on intended use and setting [N7–N9].
Organoid-derived cell therapy	GMP manufacturing and traceability; product-specific release criteria for identity, purity, potency, viability and sterility. Evaluate tumorigenicity, biodistribution, persistence, functional integration and long-term safety during development.	EU ATMP or US drug/biological-product requirements may apply. Development-stage safety studies and clinical follow-up are distinct from routine batch-release testing [N4, N12].
Therapeutic EVs	Characterize EV preparations and contaminants; establish mechanism-relevant potency and dose metrics, biodistribution, clearance, stability and manufacturing comparability.	MISEV2023 guides EV characterization and reporting but does not provide regulatory release specifications or replace GMP and product-specific validation[179].
Salivary organoid-based BoNT assays	Establish assay reproducibility, functional relevance and benchmarking against accepted neuronal potency methods.	Salivary organoid assays remain exploratory and have not been incorporated into pharmacopoeial monographs.

Requirements depend on the intended application, product characteristics and jurisdiction. Quality attributes requiring product-specific release criteria should be distinguished from developmental safety studies and long-term clinical follow-up.